



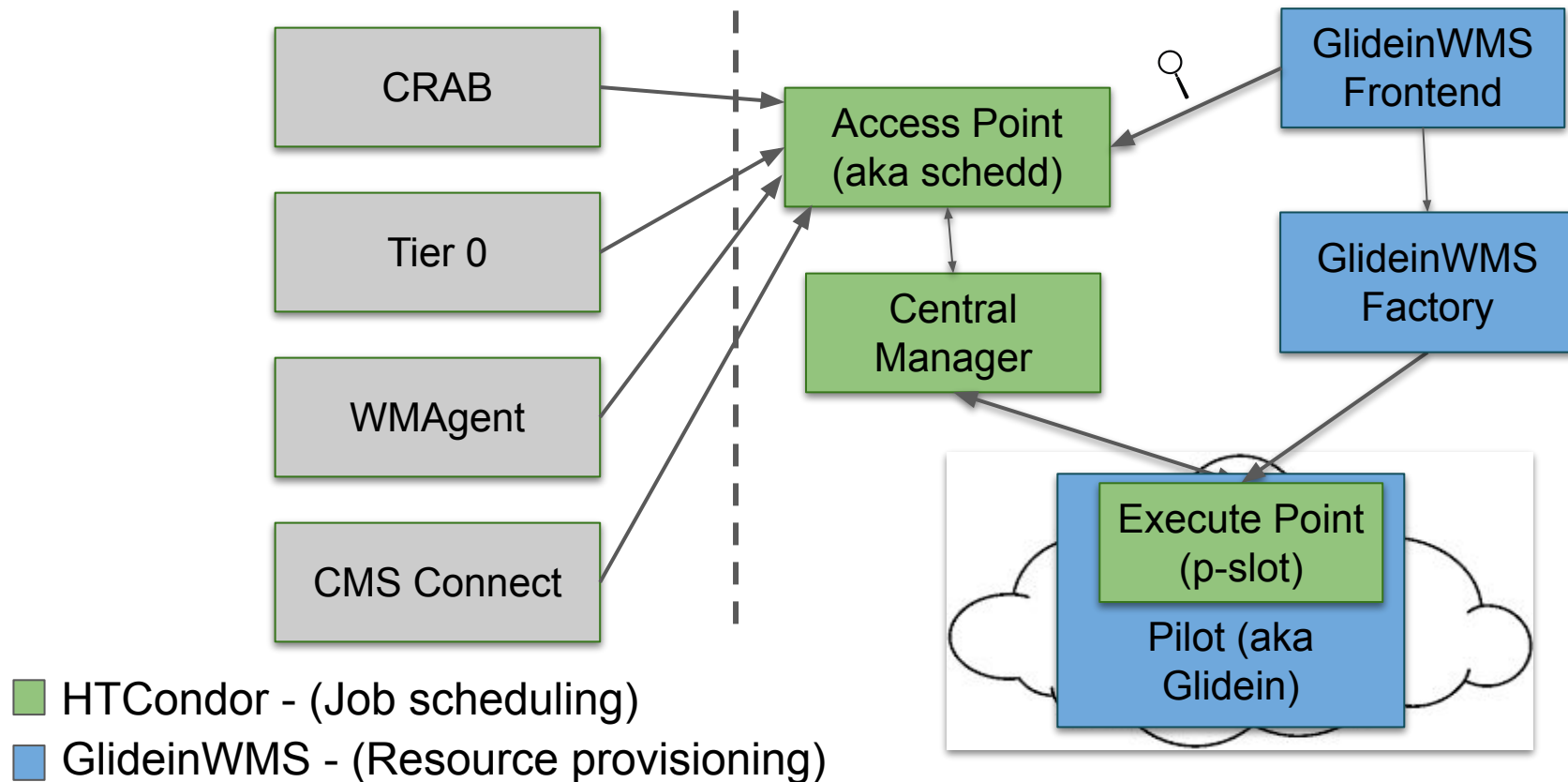
GlideinWMS: status update from development team

Marco Mascheroni, Marco Mambelli, Namratha Urs

Submission Infrastructure in WM context

Workflows

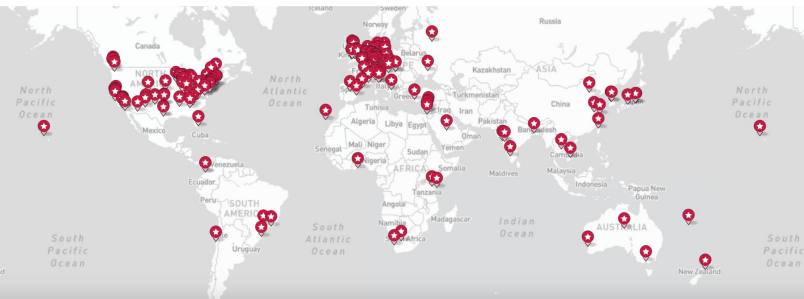
Jobs (SI)



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Map of 175 Pools Utilizing HTCSS (user reported)
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GlideinWMS and HTCondor as the backbone of computing for CMS and many other experiments

- **GlideinWMS** is used by >20 experiments without constraining development.
 - On the OSG side alone about a **dozen different experiments** and up to about **100 individual research groups** across all of science use glideinWMS.



GlideinWMS Overview (Jan–Nov 2025)



- Total commits: ~130 since January 2025 in the **stable series**
- Key **focus areas**:
 - **Factory** automation & site configuration
 - **CVMFS** and container integration
 - **Credential management** & environment handling
 - Documentation, testing, and refactoring

Active and pivotal project despite limited effort (after Bruno left, 1.35 FTE including external contributors)

Major Releases Timeline (stable series)

v3.10.9 – January 2025

- Introduced **HTCondor-distributed Apptainer** integration
- Added **JWT-authenticated log server** and improved logging
- Improved Glidein **signal handling and shutdown behavior**

v3.10.10 – January 2025

- Improved **gfdiff** tool interface and algorithm
- Added option to **check for new HTCondor tarballs**

v3.10.11 – March 2025

- Introduced **stale_age** parameter for Factory entries
- Reinstated **HTTPS monitoring** and new default token key location
- Improved **Apptainer** and **Singularity variable** handling

v3.10.14 – June 2025

- Add keyword **ALL** to **query all schedulers known to the collector**
- Added support for **Ubuntu 24** and **Alpine containers**
- Improved **CVMFS setup** and **on-demand repository configuration**
- Bug fixes for **Apptainer** environment and package conflicts

v3.10.15 – July 2025

- Enhanced **cvmfsexec mode1 (mountrepo)** with new **CVMFS_REPOS** variable
- Introduced **GLIDEIN_OVERLOAD_ENABLED**

v3.10.16 – September 2025

- New **staticextra** resource type for memory management
- Improved security: disabled **version headers** and **TRACE** in **httpd**
- Fixed environment cleanup and **cvmfsexec bugs**

v3.10.17 – November 2025

- More efficient Factory reconfig (**lower memory use, cached HTCondor tarballs**) and **EL/CentOS 10 support**
- Improved monitoring and **configuration safety** (word-checker, POSIX wrapper warnings)
- Added HTCondor 25 support (**Python bindings v2**) and fixed RRD verifier + UI issues

CMS SI
Factory Ops

And more **bugfixes** at every release! See
[CHANGELOG](#) for details

Development release 3.11

Development releases introduce **disruptive features**.

3.11.0 to 3.11.3 focusing on **credentials and generators**

- 3.11.0 and 3.11.1 introduced most of the changes, 3.11.2 and 3.11.3 have been focusing on refining features and hardening
- Designed to handle **tokens and** allow explicit choices about **credentials**
 - Distinction of **type** (SciToken, IDToken, SSH key pair) **and purpose** (site authentication, callback -ie to join the pool-, payload - used by the glidein or jobs)
 - Handling of **multiple credentials**
- Flexible **credentials management**: clear syntax to specify mandatory, optional or fallback credentials and parameters (e.g. project ID)
- Flexible runtime **generators for credentials** (SciTokens, IDTokens, custom) and parameters
 - E.g. per-site credentials, a project ID depending on the site name or gatekeeper, longer IDToken lifetime on some sites
 - This **reduces** the need for different **groups** (easier configuration for reduced operation footprint)
- The Frontend can publish self signed SciTokens (e.g. for testing, without using IAM)
- Code improvement: modern Python 3 module

3.11.4 will focus on **completing cvmfsexec support**

Next Steps and Discussion Points

Present list of **potential** projects and help us prioritizing them

Glidein Feedback Loop

- Glideins provide **runtime info** back to the Factory to **enrich configuration**
 - Useful to finetune **first matchmaking** and improve resource usage
- Potential impact of new feature:
 - GPU availability and specs => Avoid launching GPU pilots on T4 if jobs ask A100
 - Microarchitecture details => Abort jobs if you require v3 but sites only provide v2
 - Benchmark results => Could be used for splitting
 - Unprivileged singularity => List of sites that support nested containers

HPC Integration

- HPC support remains a key focus
 - Improve automation of SSH-based submissions (HEPCloud)
 - Integrate CMS 'glideins in a vacuum' into GlideinWMS
 - Containerized wrappers for easier deployment
 - Continue integration of cvmfsexec support
 - See next slides

On-demand CVMFS with GlideinWMS

How it works?

1. Check pre-requisites on the worker node
 - a. Is on-demand CVMFS requested?
 - b. Is CVMFS locally available on the worker node (CE)?
2. Determine possible cvmfsexec mode at the time of glidein startup
 - a. Can cvmfsexec mode 3 be used?
 - b. Otherwise, fall back to mode 2/1
 - c. Mount CVMFS

On-demand CVMFS with GlideinWMS

- **As of v3.10.17 (production)**

- Minimal *cvmfsexec* support (modes 1 and 2 only) available
- Automatic selection of *cvmfsexec* distribution for the worker node
- Version-based selective updating of *cvmfsexec* distributions
- Trigger-based control of (re)building *cvmfsexec* distributions

- **Upcoming in v3.11.4 (development)**

- Complete support for all three modes of *cvmfsexec*
 - Use *cvmfsexec* mode 3 when possible if worker node meets the minimum requirements
 - Fall back to *cvmfsexec* mode 1 otherwise
- Emphasis on supporting mode 3 of *cvmfsexec* to allow graceful cleanup of mount points
- Includes re-design of the Glidein and refactoring of the Glidein startup script

On-demand CVMFS with GlideinWMS: **Next Steps**

- Feature flag implementation in progress to allow flexibility to deploy/test disruptive changes for cvmfsexec
- Build the release candidate/final release RPMs for OSG
- Some known issues have previously been identified; addressing them to provide workarounds and/or fixes
 - Absence of process-level isolation with multiple Glideins [[GWMS#287](#)]
 - Disabling building of cvmfsexec distros after a previous successful build results in failure [[GWMS#305](#)]
 - Generating an el9 cvmfsexec distro on an el7 platform results in factory reconfig/upgrade to fail [[GWMS#312](#)]

Others

- Adopt **CGROUPS v2** to improve resource isolation (CPU, memory)
- Introduce Condor concurrency limits to manage high I/O workloads
 - To **cap** number of high I/O **per site**
 - **ioslots** helps protecting **worker nodes** instead
- Start using the HTCondor **native singularity invocation**
 - Requires major refactoring, but will allow to use **condor_ssh_to_job** again
- Give sites the possibility to advertise **custom attributes**
 - CMS does it through a custom validation script and SITECONF
- **Decouple Factory and schedd** components
 - Simplifies **containerized deployments** and improves flexibility

Conclusions

- GlideinWMS continues to be **actively developed**
 - Steady, **meaningful progress** in 2025
- Improvements reduce operational load, improve support for on-demand cvmfs and add new credential management
- Need to prioritize work to positions us well for future development
 - HPC integration, cgroups v2 resource control, and richer site feedback for improved provisioning
- Development resources are **limited**, prioritization matters
 - Aligning with CMS and site needs to shape the next phase of GlideinWMS

Trigger-based (Re)Build of *cvmfsexec* Distros

- Availability of a factory knob to control the (re)building of *cvmfsexec* distributions by operators
- **Goal:** Reduce the time taken to do a factory reconfig by skipping the building of distributions when not needed
- **Usage:**

```
<cvmfsexec_distro sources="" platforms="" />
```

Usage: configuration knobs in frontend and factory

(frontend)

–**GLIDEIN_USE_CVMFS** – user/jobs require cvmfs [Boolean]

(factory)

–**CVMFS_SRC** – origin from which to download CVMFS config repo [osg/egi/default]

–**GLIDEIN_CVMFS_REQUIRE** – site/factory to specify a policy about CVMFS
[required/preferred/never]